A Context-Aware Feature Fusion Method for Multi-UAV Cooperative Air Combat

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***Abstract*— Multi-UAV autonomous cooperative air warfare is an important mode of future intelligent air warfare. However, due to the complexity and uncertainty of air combat situation information, how to accurately interpret the enemy's sustained combat intent remains a major challenge. To address this problem, we propose a Context-Aware Adaptive Feature Fusion (CAAFF) method, which can effectively utilize the time-series data of battlefield situation for hierarchical feature fusion. Specifically, the input data is first subjected to dimensionality reduction processing and feature extraction by an encoder-decoder to provide high-quality low-dimensional feature representations for further feature fusion. Next, the middle layer captures attitude changes by aggregating information from neighboring nodes via a graph attention convolutional network (GACN), flexibly fusing the features of each node, and identifying complex relationships between nodes. Finally, the mechanism of stabilizing multi-attention with self-attention is used to integrate global information at the upper layer to construct an overall feature representation of the mission and realize local-to-global posture analysis. In order to enhance the interpretation of persistent operational intent, we utilize the context-aware module to construct contextual feature representations by combining current and historical state information, thus improving the depth and interpretability of mission understanding. Finally, we combine the CAAFF method with reinforcement learning and verify its performance through multiple experiments, demonstrating the applicability and effectiveness of the method in Multi-UAV cooperative air combat.**

***Index Terms*— Multi-UAV, coordinated air warfare, decision-making, context-awareness, context-aware adaptive feature fusion.**

I. Introduction

The rapid development of Artificial Intelligence (AI) technology has significantly enhanced the role of Unmanned Aerial Vehicles (UAVs) in air warfare. UAVs are widely employed in reconnaissance and surveillance, target identification and tracking [1], coordinated fire strikes, and ground support, owing to their advantages of small size, high mobility, and low cost, which indicate great potential for military applications. The multi-UAV cooperation mode is gradually becoming an indispensable component of modern air combat systems. However, the air combat environment is complex and dynamic, characterized by various challenges such as electromagnetic interference [2], adverse weather conditions, and other factors that negatively impact communication [3]. These challenges significantly limit the effectiveness of traditional ground-based station control. Therefore, realizing autonomous and cooperative operations [4] of UAVs so that they can make fast and accurate maneuvering decisions based on dynamic information is a focus of current research [5], [6].

In research addressing the UAV air combat decision-making problem, existing methods primarily include optimization techniques, game theory, expert systems, and influence diagrams. Each of these approaches has distinct advantages and disadvantages: optimization methods can identify the optimal strategy in high-dimensional spaces, but their computational complexity is high, making it challenging to meet real-time aerial combat requirements [7], [8], [9]. Game theory methods can simulate opponents' strategies to ensure the best response through Nash equilibrium; however, obtaining and computing strategic information in a multi-party dynamic environment poses significant challenges [10], [11]. Expert systems can enhance the reliability of decision-making by leveraging domain knowledge, yet they are reliant on expert input, incur high update and maintenance costs, and lack flexibility [12], [13]. Influence diagrams offer a framework for causal analysis, enabling the prediction of optimal decision-making; however, they suffer from poor real-time performance due to model complexity [14], [15]. While these methods provide valuable insights for air combat decision-making, their practical applications in multi-UAV autonomous and cooperative operations remain relatively limited.

In recent years, Reinforcement Learning, particularly Deep Reinforcement Learning (DRL), has emerged as an effective method for addressing the decision-making problem in UAV air combat. DRL integrates neural networks with reinforcement learning to optimize strategies through trial-and-error interactions between the agent and the environment [16], [17]. Various DRL algorithms have been applied to UAV air combat, including Deep Q-Network (DQN), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO). These algorithms demonstrate operational efficiency in air combat scenarios: DQN can handle decisions in discrete action spaces [18], [19], [20], DDPG enhances exploration in

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continuous action spaces through the introduction of heuristics [21], [22], [23], and PPO optimizes the tactical capabilities of multiple UAVs via centralized training and distributed execution [24], [25], [26]. However, existing network architectures still have limitations in processing complex situational information and understanding continuous operational intent in air warfare.

This paper focuses on how to effectively process complex situational information in air combat scenarios to enable UAVs to make reasonable collaborative manoeuvre decisions under the interpretation of the enemy's sustained combat intent. Compared with the existing research results, our main contributions are as follows:

- We propose an adaptive feature fusion framework based on context-awareness. Through hierarchical feature fusion of time-series situational data, the CAAFF framework not only effectively captures the complex relationships among dynamic nodes, but also enhances the context-embedded capability of the information to achieve accurate interpretation of the enemy's persistent operational intent.
- By combining this method with a reinforcement learning algorithm, the CA_MADDPG_AFF algorithm is constructed. The algorithm combines a multi-layer graph neural network to empower the intelligences to parse the dynamic battlefield environment more efficiently by adaptively processing the temporal information in the battlefield situation. Compared with traditional methods, CA_MADDPG_AFF is able to identify key tactical patterns at both local and global levels, thus optimizing the interpretation of the enemy's persistent operational intent, and improving the accuracy of strategy learning and decision-making ability in confrontation.
- Simulation experiments demonstrate the CA_MADDPG_AFF algorithm's ability to interpret the enemy's persistent combat intent and learn complex air combat maneuver strategies, confirming the applicability of the CAAFF framework in air combat missions. Comparative experiments show that the CA_MADDPG_AFF algorithm outperforms traditional methods in complex air combat scenarios.

The rest of the paper is organised as follows. Section II introduces a decision-making model for UAV air combat manoeuvres. Section III proposes a CAAFF-based RL method. Section IV details the experimental validation. Section V concludes the paper.

II. Related Works

In a multi-agent environment, the graph structure can effectively model the interaction relationships among agents. Currently, several Multi-Agent Reinforcement Learning (MARL) algorithms employ Graph Neural Networks (GNN) [27] and Graph Attention Networks (GAT) [28] for learning representations of agent relationships.

For example, one study utilized hierarchical heterogeneous graphs to model the hierarchical relationships among multiple heterogeneous agents and constructed a hierarchical graph attention network for representation learning [29]. Another study employed a complete graph to model the relationships between agents and proposed a novel game abstraction mechanism based on a two-stage attention network (G2ANet). This mechanism not only indicates whether there is an interaction between two agents but also assesses the significance of that interaction [30]. Additionally, a study decomposed the joint team strategy into a graph generator and a graph-based coordination strategy to facilitate efficient coordination among agents. The graph generator uses an encoder-decoder framework that outputs directed acyclic graphs (DAG) to capture the underlying dynamic decision structure, while DAG constraints and depth-constrained optimization are applied to balance efficiency and performance [31]. Furthermore, the proposed GPMF algorithm combines graph attention networks with partially observable mean field theory to address the local optimum problem arising from insufficient consideration of neighboring feature information. GPMF utilizes a graph attention module and a mean field module to characterize how agents are influenced by the actions of other agents at each time step. The graph attention module comprises a graph attention encoder and a differentiable attention mechanism that generates dynamic graphs to represent the relationships among agents [32]. Finally, an algorithm called GQN is proposed, which employs a graph-based approach to construct node and edge features that convert potential relationships among agents into graph representations, providing a robust relationship induction bias. Graph network blocks are structured to enable multi-layer graph messaging, abstracting communication between agents into information aggregation at graph nodes [33]. However, the above studies failed to explore time-series based posture relationships. Relying only on the state information at a certain moment in time cannot fully depict the whole picture of the battlefield, and it is even more difficult to deeply understand the enemy's continuous combat intent.

III. Decision-Making Model for UAV Air Combat Maneuvers

In this section, we describe the process of air combat scenarios of UAV swarms and explain the air combat manoeuvre decision-making model for UAVs, including the UAV's motion model, the air combat posture assessment model, the state space and the action space. Since the second and third sections of the article contain many formulas and symbols, we have added a glossary to provide a description of some of the symbols, as shown in Table I.

A. Scene Description

The UAV air combat scenario is set up as follows: two UAVs meet in a certain airspace, first carry out over-the-horizon reconnaissance and long-range missile attack, and enter the close-range high-mobility confrontation stage after some UAVs break through the missile blockade. The goal of both sides is to cooperate in destroying all of the other side's drones. Both the red and blue males are isomorphic. Each drone can only detect drones within its observation range. The conical area in front of each drone at a certain angle

TABLE I
GLOSSARY

Notation	Describable
X_{ij}	The relative posture value
s_p^{ij}	The relative position vector
s_v^{ij}	the relative velocity vector
s_i^j	State vector
$H^{(l)}$	The node feature matrix of layer l
α_{ij}	Attention coefficients
H'	Mid-level feature matrix
g_t^*	Global Feature Vector
f_{MSG}	Message function
f_{AGG}	Aggregation function
TE	Time-coded function
t_i'	Timestamp
h_i^E	Reconstructed observations

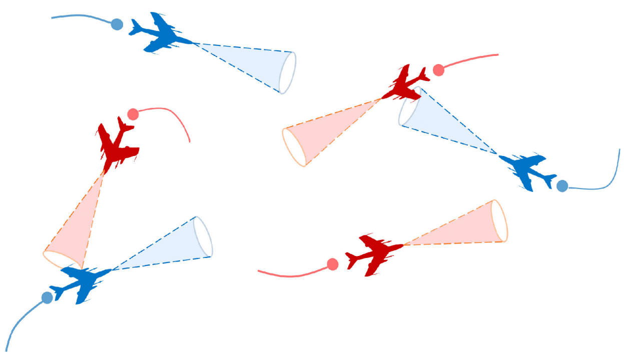


Fig. 1. UAV air combat scenario.

for a certain distance is the attack zone of that drone, and the conical area behind each drone at a certain angle for a certain distance is the kill zone of that drone, as shown in Fig. 1. When a UAV is within the attack zone of an enemy UAV and does not escape from the attack zone within a certain period of time, the UAV will be destroyed. The red UAVs are controlled by scripting and the blue UAVs are controlled by RL method. The blue drones need to collaborate with each other to intelligently make the best maneuvering decisions based on their observations of other drones' information. Blue-side drones need to ensure their own survival while destroying the enemy.

B. UAV Motion Model

Since the maneuvering decision of the UAV mainly considers the relative position and relative velocity relationship between UAVs. Therefore, this paper adopts the three-degree-of-freedom model as the motion model of the UAV, as shown in Fig. 2 To simplify the problem, it is assumed that the velocity direction of the UAV coincides with the fuselage axis, and the motion model of the UAV is shown in Equation (1).

$$\begin{cases} \dot{x} = v\cos\theta\sin\varphi \\ \dot{y} = v\cos\theta\cos\varphi \\ \dot{z} = v\sin\theta \end{cases} \tag{1}$$

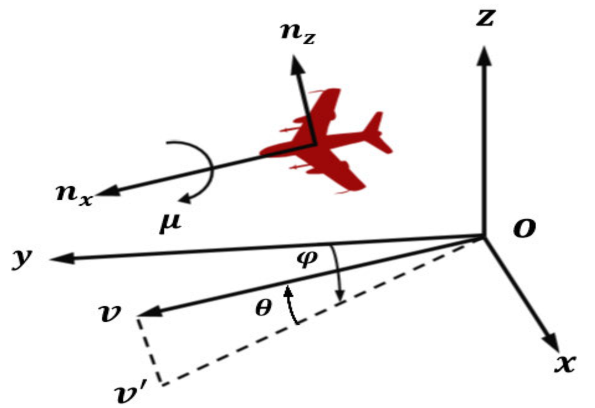


Fig. 2. Motion model of UAVs.

where x, y, z are the UAV position coordinate values, $p = [x, y, z]$, v is the velocity vector of the UAV, $\dot{x}, \dot{y}, \dot{z}$ are the rate of change of the position in the three directions indicated by v. μ, θ and ϕ represent the roll angle, trajectory angle and heading angle respectively. $\dot{v}$ is the projection of v on the xoy plane. The dynamics model of the UAV is shown in Equation (2)

$$\begin{cases} \dot{v} = (n_x - \sin\theta)g \\ \dot{\gamma} = \dfrac{(n_z\cos\mu - \cos\theta)g}{v} \\ \dot{\psi} = \dfrac{n_z g\sin\mu}{v\cos\theta} \end{cases} \tag{2}$$

where g is the gravitational acceleration, n_x is the overload in the velocity direction, n_z is the overload in the normal direction and μ is the roll angle. The basic control parameters n_x, n_z and μ in the kinematic model can be represented as a control input vector $\mathbf{u} = [n_x, n_z, \mu]$, which controls the velocity of the UAV, and n_z and μ control the direction of the velocity vector.

C. Posture Assessment Model

The air combat posture assessment between two homogeneity and isomorphism UAVs is mainly related to factors such as relative angle and relative distance. The relationship between the enemy (blue) and our (red) sides in the battle is shown in Fig. 3, $\alpha_r$is called the angle of attack, which can be expressed by Equation (3). We will attack the enemy only when the enemy is in our attack range. Equation (4) can be used to describe whether the attack condition is reached or not.

$$\alpha_r = \arccos\frac{D_{\text{r}}^{\text{b}}\cdot \text{v}_r}{\|D_{\text{r}}^{\text{b}}\|\cdot\|\text{v}_r\|} \tag{3}$$

$$\begin{cases} \alpha_r \leq \alpha_{\max} \\ D_{att}^{\min} \leq\| D_{\text{r}}^{\text{b}} \|\leq D_{att}^{\max} \end{cases} \tag{4}$$

where $D_{\text{att}}^{\min}$ and $D_{\text{att}}^{\max}$ are the shortest and longest attack distances. $\alpha_{\max}$ is the maximum attack angle, which is set to 45°. α_b is the escape angle which can be expressed by Equation (5). In the confrontation, our side not only wants to keep the blue color within its attack range but also tries to avoid being attacked by the enemy. So our advantageous

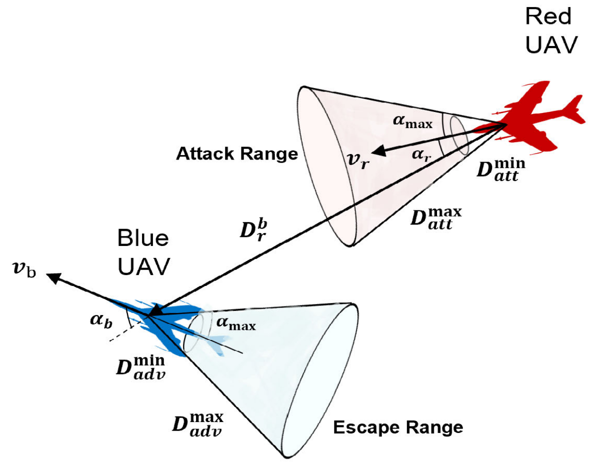


Fig. 3. The situation of the enemy and us in the air battle.

area is defined behind the enemy, which can be expressed by Equation (6).

$$\alpha_b = \arccos \frac{D_{\mathrm{r}}^{\mathrm{b}} \cdot \mathrm{v}_b}{\|D_{\mathrm{r}}^{\mathrm{b}}\| \cdot \|\mathrm{v}_b\|} \tag{5}$$

$$\begin{cases} \alpha_b \le \alpha_{esp} \\ D_{adv}^{\min} \le \| D_{\mathrm{r}}^{\mathrm{b}} \| \le D_{adv}^{\max} \end{cases} \tag{6}$$

where $D_{\mathrm{adv}}^{\min}$ and $D_{\mathrm{adv}}^{\max}$ are the minimum and maximum dominance distances and α_{esp} is the maximum escape angle of red in its dominance region. In summary, the optimal attack conditions for red UAV can be described by Equation (7).

$$\begin{cases} \alpha_b \le \alpha_{esp} \\ \alpha_r \le \alpha_{\max} \\ \|\mathrm{v_b}\| \le \|\mathrm{v_r}\| \\ D_{att}^{\min} \le \| D_{\mathrm{r}}^{\mathrm{b}} \| \le D_{att}^{\max} \end{cases} \tag{7}$$

According to the relative posture assessment model, the relative posture function is set by considering the relative angle advantage and relative distance advantage comprehensively, as shown in Equation (8). ω is the weight of the relative angle advantage and the relative distance advantage $\omega_1 = 0.3$, $\omega_2 = 0.3$, $\omega_3 = 0.2$, $\omega_4 = 0.2$.

$$\mathrm{X}_{ij} = \omega_1 \alpha_r + \omega_2 \alpha_b + \omega_3 D_{\mathrm{r}}^{\mathrm{b}} + \omega_4 (\|\mathrm{v_r}\| - \|\mathrm{v}_b\|) \tag{8}$$

X_{ij} is the relative posture value between agent i and j. Each time, the enemy with the largest relative posture value is selected as the combat object based on the matrix of posture function values.

D. Joint State Space

We use the state space of the UAV air combat maneuver decision model to describe the air combat situation. A many-to-many form of air combat is different from a one-to-one scenario. Each UAV needs to consider the states of multiple other UAVs including the enemy within its sensing range and the friend with cooperative relationship. So the state space of a UAV should consist of the state vectors of multiple sensed UAVs. When UAVi is the subject-object, the relative position vector and the relative velocity vector of UAVj under the coordinate system of UAVi constitute the state vector of a single UAV $s_i^j = [s_p^{ij}, s_v^{ij}, \mathrm{X}_{ij}]$, where $s_p^{ij} = [x_i^j, y_i^j, z_i^j, D_i^j]$ is the relative position vector, $s_v^{ij} = [v_x^i, v_y^i, v_z^i, \alpha, \alpha_b]$ is the relative velocity vector, and X_{ij} is the relative situation value. For the subject UAV i, the relative state of the enemy UAVm is denoted as s_i^m and the relative state of the friendly UAVn is denoted as s_i^n. Then the state space of UAVi can be expressed by Equation (9).

$$S_i = [\mathrm{U}s_i^{\mathrm{m}} | D_i^{\mathrm{m}} \le D_{obs}^{\max}, \mathrm{U}s_i^{\mathrm{n}} | D_i^{\mathrm{n}} \le D_{obs}^{\max} (i \ne \mathrm{n})] \tag{9}$$

where D_{ij}^{m} is the relative distance between UAVi and UAVm, and $D_{obs}^{\max}$ is the maximum sensing distance of the sensors of the UAV.

TABLE II
THE BASIC ACTIONS VALUES IN THE DESIGNED ACTION SPACE

NO	Maneuver	n_x	n_z	μ
Action1-3	Forward	-1, 0, 1	1	0
Action4-6	Right turn	-1, 0, 1	3.5	$-\arccos(2/7)$
Action7-9	Left turn	-1, 0, 1	3.5	$\arccos(2/7)$
Action10-12	Downward	-1, 0, 1	3.5	π
Action13-15	Upward	-1, 0, 1	3.5	0
Action16-18	Downward and left turn	-1, 0, 1	3.5	$3\pi/4$
Action19-21	Downward and right turn	-1, 0, 1	3.5	$-3\pi/4$
Action22-24	Upward and left turn	-1, 0, 1	3.5	$\pi/4$
Action25-27	Upward and right turn	-1, 0, 1	3.5	$-\pi/4$

E. Joint Action Space

The action space for UAV maneuver decision-making is the UAV maneuver library. The UAV air combat maneuver library can be divided into two sub-banks: the typical tactical maneuver library and the basic maneuver library. Typical tactical maneuver library includes tactics such as the Cobra maneuver and Yoyo maneuver. However, it is complicated and difficult to establish a tactical maneuver library, so the basic maneuver library proposed by NASA [34], [35] is usually adopted as the maneuver library of UAVs in the design of maneuver space. However, these seven maneuvers do not well reflect the maneuvers that need to be made in air combat. The aircraft can be categorized into uniform flight, accelerated flight, and decelerated flight by speed control; straight flight, left turn, and right turn by roll angle control; and straight flight, climb, and descent by pitch angle control. The combination of these three yields 27 maneuvers, and based on the UAV motion model, $[n_x, n_z, \mu]$ is used to construct the action space for air combat maneuver decision-making. The UAV maneuver action library is shown in Table II.

IV. Reinforcement Learning Based Decision Making Strategy for Air Combat Maneuvers

In order to be able to effectively process the complex situational information of the battlefield and decipher the enemy's persistent operational intent, we hierarchically process the situational information of UAV swarms through a multi-level

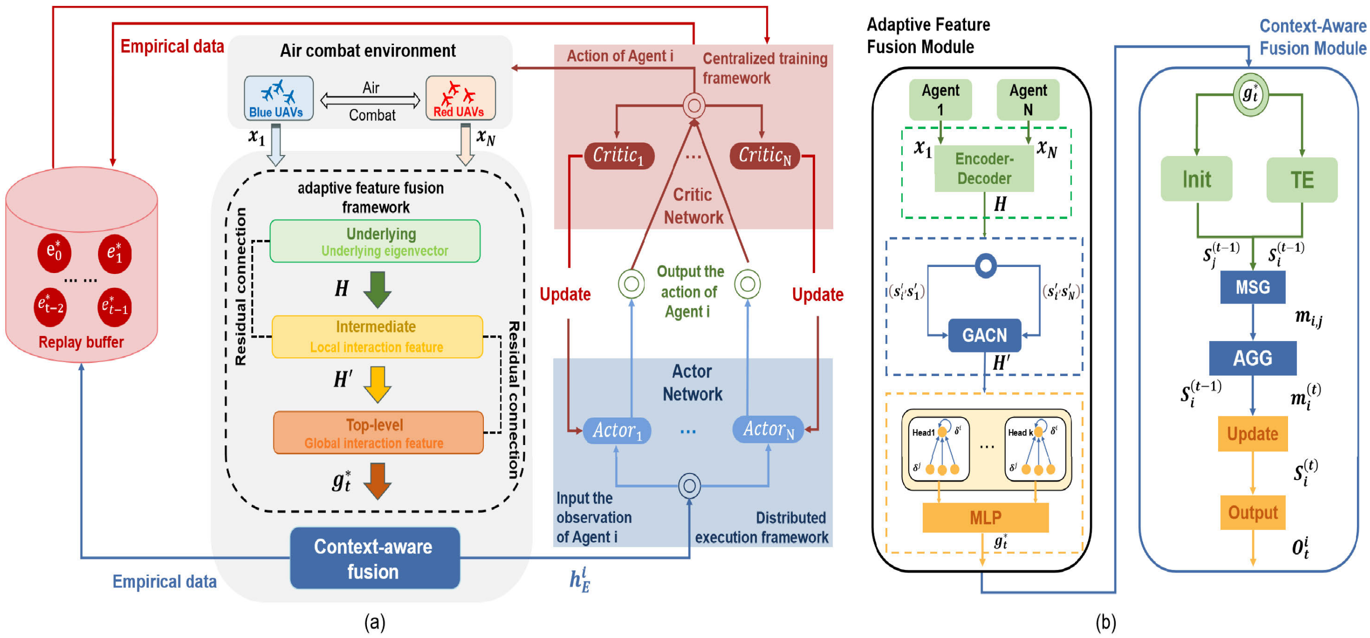


Fig. 4. (a) RL-based CAAFF frame diagram; (b) CAAFF framework details.

adaptive feature fusion framework. The context-aware module is utilised to interpret the enemy's persistent operational intent through the time-series information data [36]. Fig. 4 (a) shows the implementation of combining the RL approach with the CAAFF framework. Fig. 4 (b) shows the technical details of the framework.

A. Context-Aware Multi-Level Adaptive Feature Fusion Framework

In this section, the CAAFF framework is described in terms of a multi-level adaptive feature fusion module, context-aware module, etc. The specific technical details of the framework are shown in Fig. 4(b).

1) Multi-Level Adaptive Feature Fusion Module: The UAV passes the state information captured by the sensors to the underlying encoder-decoder, which is responsible for extracting low-dimensional key features from the complex sensor data. The encoder-decoder formulation is shown in (10).

$$\begin{cases} z_i = ReLU(W_{enc}x_i + b_{enc}) \\ S_i = ReLU(W_{dec}z_i + b_{dec}) \\ \mathcal{L}(x_i, S_i) = \|x_i - S_i\|^2 \end{cases} \tag{10}$$

where z_i is the feature vector of UAV$_i$ extracted after dimensionality reduction by the encoder, S_i is the feature vector of UAV$_i$ reconstructed by the decoder, and $\mathcal{L}$ is the loss function of the encoder-decoder expressed in terms of the mean-square error. W is the weight matrix, b is the bias term, and ReLU is the activation function.

The node feature matrices extracted by the encoder-decoder dimensionality reduction are fed into a Graph Attention Convolutional Network (GACN), which is used to model complex situational relationships between UAVs.

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{11}$$

where $H^{(l)}$ denotes the node feature matrix of layer l. Initial inputs $H^{(0)} = [S_1, S_2, \ldots, S_N]^T$. $\tilde{A}$ is a self-joining adjacency matrix $\tilde{A} = A + I_N$, A is the adjacency matrix that represents the structural relationships of the graph at that moment in time. I_N is the unit matrix that is used to contain each node's own features. $\tilde{D}$ is the degree matrix of $\tilde{A}$, $W^{(l)}$ is the weight matrix of layer l, and σ is the activation function ReLU function.

The graph attention mechanism can adaptively fuse the feature information of different nodes and flexibly learn the complex relationship between nodes. The computational formulas are shown in (12)-(13).

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}(\beta^\top[\mathbf{W}h_i\|\mathbf{W}h_j])\right)}{\sum_{k\in\mathcal{N}(i)}\exp\left(\text{LeakyReLU}(\beta^\top[\mathbf{W}h_i\|\mathbf{W}h_k])\right)},$$
$$\sum_{k\in\mathcal{N}(i)} \alpha_{ij} = 1 \tag{12}$$

where W is the weight matrix which is used to transform the feature vector of each node to a higher dimensional space. β is the parameter vector used to compute the attention coefficient. $\|$ denotes the splicing operation and LeakyReLU is the activation function that is used to increase the nonlinearity of the network. h_i is the feature vector of UAVi. $\mathcal{N}(i)$ denotes the set of neighbors of node i.

In order to improve the expressive power of the model and learn more complex feature representations. We repeat the above process several times (each time using different sets of parameters W and *beat*), and then combine the features output from each head in a spliced or averaged fashion. The final node features can be represented as:

$$h_i = \|_{k=1}^{K}\sigma\left(\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^k\mathbf{W}^k h_j\right) \tag{13}$$

where $K = 4$ is the number of attention heads, α_{ij}^k and W^k are the attention coefficients and weight matrices for the k-th attention head. The node features are fed into Gumbel Softmax to obtain the mid-level feature matrix H'

The high-level level, on the other hand, employs multi-head attention to stabilize the self-attention to integrate the information from all UAV networks and construct the global task vector that represents the overall task. The formulas for the high level of the framework are shown in (14)-(16).

$$Q = H'W^Q, \quad K = H'W^K, \quad V = H'W^V \tag{14}$$

where H' is the input feature matrix, and W^Q, W^K and W^V are the weight matrices corresponding to the query, key and value respectively.

$$\text{head}_i = \text{Attention}(Q, K, V)V = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{15}$$

The output obtained from each header computation is the attention weights, d_k is the dimension of the key vector, and the division operation is used to scale the result of the dot product, which contributes to the stability of the training. softmax is a normalisation function that normalises the relevance scores of queries and keys into a probability distribution, which is used to generate attention weights to determine how much attention the model pays to different pieces of information. Finally, the outputs from all heads are stitched together and fused by a final linear layer.

$$g_t^* = \text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_l)W' \tag{16}$$

W' is another weight matrix that is used to fuse the outputs of all the heads into a final output vector.

2) Context-Aware Modules: Due to the highly time-varying nature of the air combat environment and the situational relationships between UAVs, relying solely on the state information at a given moment in time is insufficient to portray the entire battlefield. In order to capture the time-series based situational relationships, so as to better interpret the enemy's persistent combat intent. In this paper, we propose a context-aware module based on time-series graphical neural networks. It can combine the global information of the current moment and the historical moment, construct embedding vectors based on contextual information, and achieve the reconstruction of the original observation data, so as to better guide UAVs to make decisions on air combat manoeuvres. The calculation formulas of the context-aware module are shown in (17)-(21). Information generation is calculated by equation (17) and information aggregation is calculated by equation (18).

$$m_{ij}(t) = f_{\text{MSG}}\left(s_i^{(t-1)}, s_j^{(t-1)}, e_{ij}, t_{ij}\right) \tag{17}$$

$$m_i^{(t)} = f_{\text{AGG}}\Big(m_{ij}(t) | j \in N(i)\Big) \tag{18}$$

where e_{ij} is the characteristics of the edge ij, t_{ij} is the time when the edge occurs. $s_i^{(t-1)}$ is the state of the previous moment, the initial inputs $s_i^{(o)} = f_{\text{init}}(h_i')$. h_i' are the initial characteristics of the node, f_{init} is the initialization function. f_{MSG} is the function used for generating the message. $\mathcal{N}(i)$ denotes the neighbors of the node and f_{AGG} is the aggregation function. The computational formula used to update the node features is shown in Equation (19)

$$s_i^{(t)} = f_{\text{update}}\left(s_i^{(t-1)}, m_i^{(t)}\right) \tag{19}$$

where $\mathbf{m}_i^{(t)}$ is the message received at the current moment and $\mathbf{f}_{\text{undate}}$ is the state update function. The encoding based on temporal features is given by Equation (20)

$$t_i' = TE(t_i) \tag{20}$$

where t_i is the timestamp and TE is a time-coded function, which we choose to repre.sent as a cosine function with periodicity and continuity. Its output range is fixed within $[-1, 1]$, which helps to avoid the problem of numerical instability during model training. The final output observation is given by Equation (21)

$$h_i^E = f_{\text{output}}\left(s_i^{(t-1)}, m_i^{(t)}, t_i'\right) \tag{21}$$

where f_{output} is the output function, h_i^E is the observation vector of the output.

B. Reinforcement Learning Algorithms

This paper proposes the CA_MADDPG_AFF algorithm based on the MADDPG [37] framework. The MADDPG algorithm uses the off-line mode [38] for centralized training and the on-line mode for distributed execution. In the off-line mode, the experience of all agents can be centralized for training, which helps to learn more complex cooperative strategies and globally optimal solutions in the entire system. In the on-line mode, each agent can adjust its strategy according to the current environment and the real-time dynamics of other agents. The shortcomings of the MADDPG algorithm are poor convergence performance and poor optimization performance. Therefore, we use the CAAFF framework to generate embedding vectors instead of the original observation value input of the critic and actor networks. The embedding vector of each agent summarizes the relationship between the agent and the other agents and the environment. The calculated embedding vector of each agent is then used to calculate the Q value and the action in actor-critic. Therefore, the method of calculating the Q value and action in MADDPG is updated to Equations (22) and (23).

$$\begin{aligned} Q_i(o_i, a_i) &\iff Q(h_i^E, a_i; \phi_i), a_i = \mu(h_i^o; \theta_i) \\ &\iff \mu(h_i^E; \theta_i) \end{aligned} \tag{22}$$

The loss function is calculated as follows:

$$\begin{aligned} \mathcal{L}(\phi_i) &= \mathbb{E}_{o,a,r,o'}\left[\left(Q_i'(h_i^E, a_i; \phi_i) - y_i\right)^2\right], \\ y_i &= r_i + \gamma Q_i'\left(h_i^E, a_i; \phi_i\right)\Big|_{a_i = \mu'(h_i^E; \theta_i')} \end{aligned} \tag{23}$$

where Q_i' and μ are the target critics and actor networks for stable learning using the delay parameter, respectively. The gradient is computed as follows:

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{o,a\sim D}\left[\nabla_{\theta_i}\mu(h_i^E; \theta_i)\nabla_{a_i} Q_i'(h_i^E, a_i; \phi_i)\Big|_{a_i = \mu(h_i^E; \theta_i)}\right] \tag{24}$$

where D is the empirical buffer and θ_i and ϕ_i are network parameters.

Algorithm 1 CA_MADDPG_AFF Training Procedure For N-to-N UAV Swarm Air Combat

1. Set n UAVs, the number of training sessions num_episode, and the maximum number of iterations per session max_episode_size. $O(1)$
2. Initialize buffer D, policy network parameters $\pi(s|\theta)$, value network parameters $Q(s,a|\theta)$, and CAAFF frame parameter θ. $O(1)$
3. **for** episode=1 to num_episode do. $O(num\_episode)$
4. Initialize the environment to generate both drones at random locations.
5. **for** t=1 to max_episode_size do. $O(max\_episode\_size)$
6. **for** each UAV i, input the original observations s_i into the CAAFF framework o_i. $O(n)$
7. Perform actions $(a_1,\ldots,a_N)$ and earn rewards R and new status s'.
8. Collect (s,a,r,s') of N UAVs into N independent trajectories in buffer D.
9. Set $s \leftarrow s'$.
10. If episode terminates then.
11. break.
12. **end for**
13. **for** each UAV $i=1$ to n do. $O(n)$
14. Sample S samples (s,a,r,s') randomly from the experience pool D.
15. Compute $y_i = r_i + \gamma Q_i'(h_i^E, a_i; \phi_i)|_{a_i=\mu'(h_i^E;\theta_i')}$
16. Update evaluation function: $L(\phi_i) = \mathbb{E}_{o,a,r,o'\sim D}[(Q_i'(h_i^E, a_i;\phi_i) - y_i)^2]$
17. Gradient update action function: $\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{o,a\sim D}[\nabla_{\theta_i}\mu(h_i^E;\theta_i)\nabla_{a_i} Q_i'(h_i^E, a_i;\phi_i)|_{a_i=\mu(h_i^E;\theta_i)}]$
18. **end for**
19. **end for**
20. **end for**

C. Reward Function

Reinforcement learning algorithms are mainly driven by rewards, constructing reward functions through task requirements and shaping rewards to guide UAVs to learn optimal strategies. Therefore, it is necessary to design the reward function that best fits the actual task to guide the learning process. The reward function of this algorithm is divided into the following four parts:

1) Relative Situational Rewards R_s: The first part of the reward is the relative situation reward reward. The matrix of posture function values is computed once at each step. For each UAV at each moment in time, the relative posture value of that UAV relative to other enemy UAVs in its observation range is calculated. The relative posture function is multiplied by the reward factor as the relative posture reward value. The relative posture reward of the UAV is calculated by Equation (25)

$$\mathrm{R}_s = \begin{cases} \gamma_S \cdot \mathrm{X}_{i,j} & , \exists UAV_{\mathrm{j}}, D_{\mathrm{i}}^{\mathrm{j}} \le D_{\mathrm{i}}^{\mathrm{o}} \\ 0 & , \forall UAV_{\mathrm{j}}, D_{\mathrm{i}}^{\mathrm{j}} > D_{\mathrm{i}}^{\mathrm{o}} \end{cases} \quad (25)$$

where X_{ij} denotes the posture function of UAV i relative to UAV j, which can be calculated from equation (8). D_i^j is the distance between UAV i and UAV j, D_i^0 is the maximum sensing distance of the sensor, and γ_s is the reward coefficient of the relative posture reward.

2) Out-of-Bounds Penalty R_p: The second part of the reward is the out-of-bounds penalty. The out-of-bounds penalty can guide the UAV to make maneuvering decisions within a reasonable air combat zone. The out-of-bounds penalty function can be expressed by Equation (26)

$$\mathrm{R_p} = \begin{cases} -\gamma_{\mathrm{p}} & , \mathrm{pos(i)} > Zone_{\mathrm{high}}\ or\ \mathrm{pos(i)} < Zone_{\mathrm{low}} \\ 0 & , Zone_{\mathrm{high}} \ge \mathrm{pos(i)} \ge Zone_{\mathrm{low}} \end{cases} \quad (26)$$

where Z_{high} is the maximum value of the battlefield boundary, Z_{low} is the minimum value of the battlefield boundary, and γ_p is the reward factor for the transgression penalty.

3) Rewards for Defeat R_d: The third part of the reward is the defeat reward, which is given to our UAV whenever it defeats the enemy UAV and vice versa. The defeat reward function can be expressed by Equation (27)

$$R_d = \begin{cases} \gamma_d & , \text{Blue defeat Red} \\ -\gamma_d & , \text{Red defeat Blue} \\ 0 & , \text{else} \end{cases} \quad (27)$$

4) Global Rewards R_w: The fourth part of the reward is the global reward, which is determined based on the group win or loss of the drone air battle. Rewards are determined at the end of each episode. Depending on the number of drones remaining on each side at the end of the confrontation, the same global reward or penalty is given to all our drones. This can lead drones to learn cooperative group maneuvering strategies and eventually win air battles. The formula for calculating the global reward can be experssed by Equation (28)

$$R_{\mathrm{w}} = \begin{cases} \gamma_{\mathrm{w}} & , Number_{\mathrm{blue}} > Number_{\mathrm{red}} \\ -\gamma_{\mathrm{w}} & , Number_{\mathrm{blue}} < Number_{\mathrm{red}} \\ 0 & , Number_{\mathrm{blue}} = Number_{\mathrm{red}} \end{cases} \quad (28)$$

where $\mathrm{Number_{blue}}$ and $\mathrm{Number_{red}}$ are the number of remaining drones of both sides at the end of each episode.

5) Combined Rewards R: The combined reward consists of a relative situational bonus, a out-of-bounds penalty, a defeat bonus and a global bonus, as represented by Equation (29).

$$\mathrm{R} = \begin{cases} R_{\mathrm{s}} + R_{\mathrm{p}} + R_{\mathrm{d}} & , \text{game is not done} \\ R_{\mathrm{s}} + R_{\mathrm{p}} + R_{\mathrm{d}} + R_{\mathrm{w}} & , \text{game is done} \end{cases} \quad (29)$$

The reward coefficients involved in the rewards are assigned as $\gamma_s = 0.9$, $\gamma_p = 20.0$, $\gamma_d = 50.0$ and $\gamma_w = 100.0$.

V. Experimentation

A. Experimental Design

We verified the effectiveness of the proposed method through simulation experiments. Two simulation scenarios of UAV air combat were constructed according to the different maneuver scripts executed by the enemy. In the first scenario, the enemy executes a uniform straight-line flight strategy (dashed line), and our side confronts it with the models trained by CA_MADDPG_AFF, MADDPG_AFF, and CA_MADDPG algorithms (solid line), respectively. The following confrontation scenarios under three initial superior and inferior positions are considered, respectively:

1) Face-to-Face: The two aircraft are initially facing each other. Both sides approach and then maneuver against each other. This scenario can test the flexibility and maneuvering ability in a head-on confrontation.

TABLE III
UAV ATTRIBUTE SETTINGS IN SIMULATION EXPERIMENTS

Describable	Value
Initial flight speed	150m/s
Maximum flight speed	250m/s
Minimum flight speed	100m/s
Maximum Observation Distance	[10×10×10]km
Maximum Attack Distance	500m
Maximum dominant distance	500m
Maximum direction angle	45°
Range of heading angle and pitch angle	$(-\pi/2, \pi/2)$

2) Back-to-Back: The two aircraft start out back-to-back. Both sides reach an advantageous attack position through maneuvering, this scenario can test the overall combat performance.

3) Advantageous and Disadvantageous Positions: our side is in a disadvantageous position at the beginning, and the enemy is in an advantageous position, this scenario can test the ability of defense and counterattack in the face of danger.

In the second scenario, we use the CA_MADDPG_AFF algorithm to fight against the baseline algorithm MADDPG and to more closely resemble the real scenario, the initial positions of both UAVs are randomly generated in a reasonable area for group confrontation. This scenario better reflects the performance and effectiveness of the algorithm.

In the UAV swarm air combat scenario, both UAVs move in a 3D airspace of 10000m. It is assumed that the UAVs of both sides have the same performance, as shown in Table III. The simulation experiment environment is Windows 11 operating system, Intel i5-13400F processor and NVIDIA RTX 4060 graphics card.

B. Simulation Experiment

As shown in Fig. 5, we plot the combat trajectory in the air combat scenario to better demonstrate the effect of the algorithm. We draw airplane models at the start and end positions to show the start and end situations. In addition, to show more details of the battle, we divide the trajectory into several equal parts and label them with corresponding numbers. Two trajectories with the same number correspond to the same time step. In Fig. 5, trajectories of the same color represent the same battle, with solid trajectories representing our side and dashed trajectories representing the opponent. The result of each battle is labeled at the ending position.

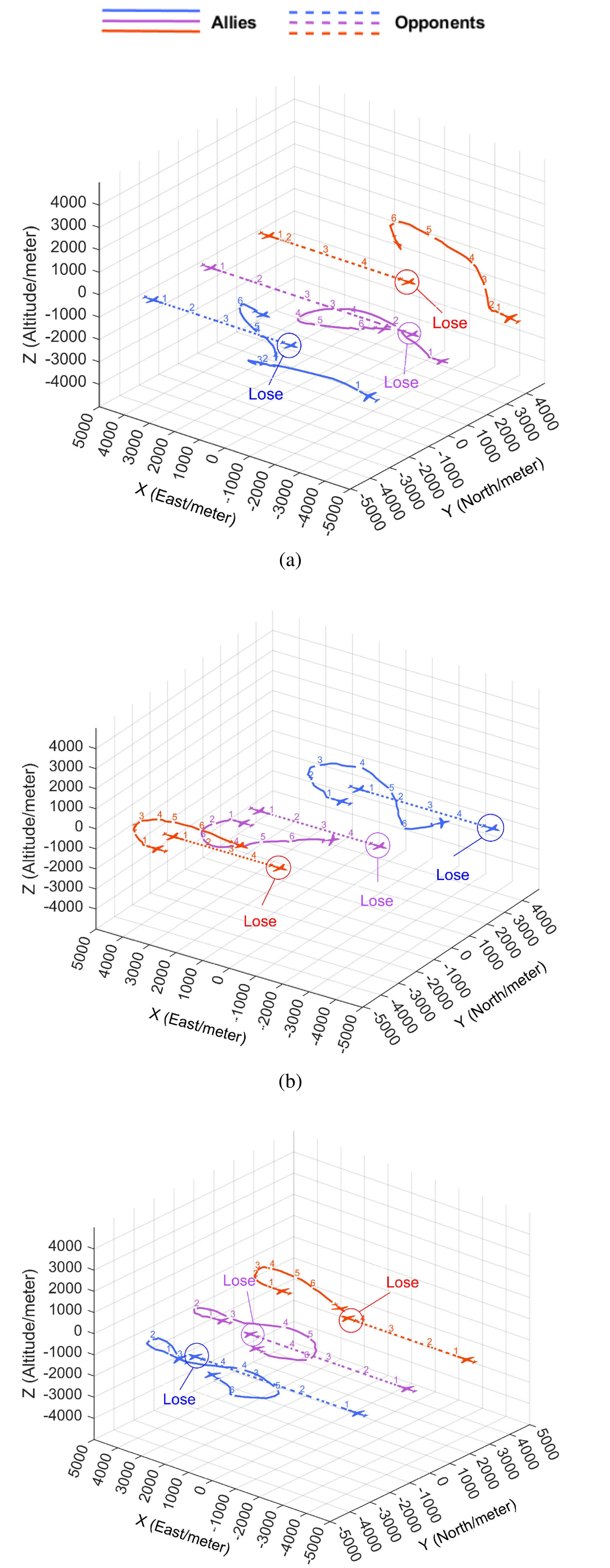


Fig. 5. Combat trajectories of models trained with CA_MADDPG, MADDPG_AFF, and CA_MADDPG_AFF algorithms under different advantage and disadvantage positions against a uniform straight flight strategy.

Fig. 5(a)(b)(c) shows the 1v1 battle trajectory executed by the algorithm for the horizontal straight flight strategy. As shown in Fig. 5(a), the initial positions of the two sides are face-to-face, and after both sides approach quickly, we use long-range maneuvering to adjust our position and angle to avoid a face-to-face attack while still being able to maneuver to the attacking area of superiority to defeat the enemy. In Fig. 5(b), the initial positions of the two sides are back-to-back. After the battle starts, we turn the direction by pulling upward and quickly chase the enemy from the back, and successfully defeat the enemy after reaching the attack conditions. In Fig. 5(c), we start the game in a disadvantageous position, and during the enemy's pursuit, we quickly carry out defense and counter-attack, avoiding the attack and at the same time being able to occupy an advantageous position to defeat the enemy.

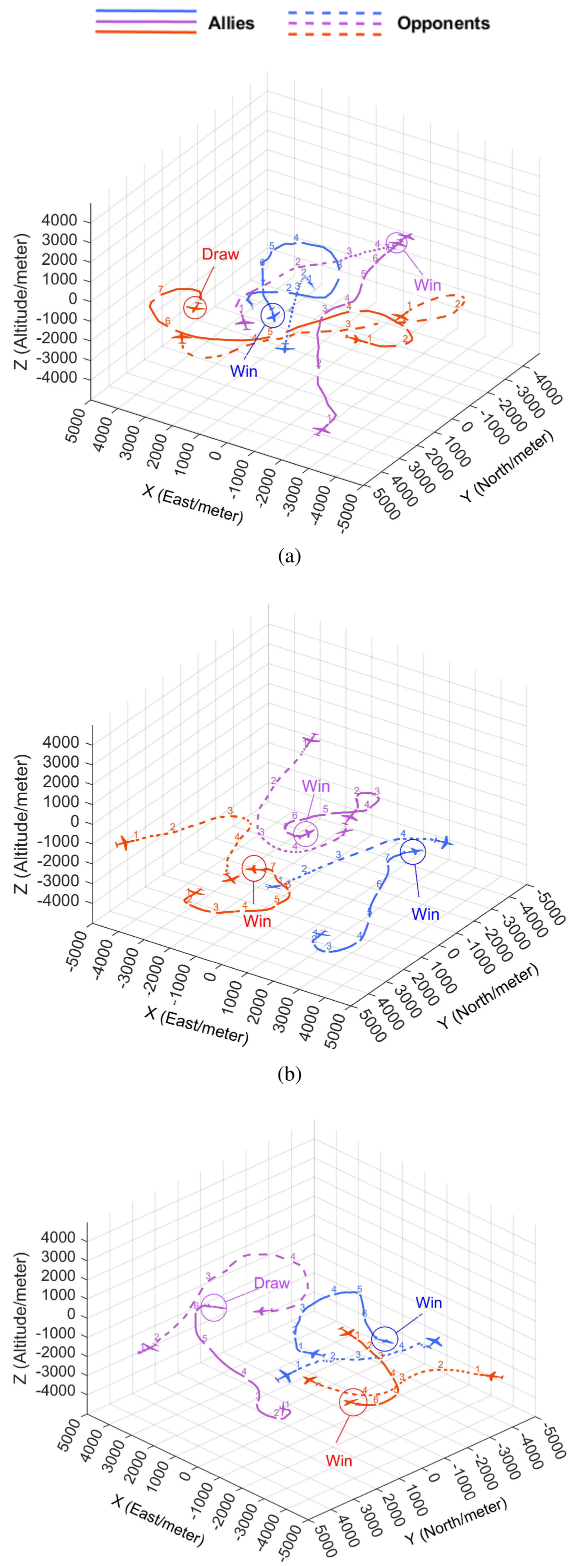
It is also easy to see from the Fig. 5 that although all of them can successfully destroy the enemy, the efficiency of the maneuver strategies adopted by the algorithms applying different modules varies. The CA_MADDPG algorithm, which only uses the context-aware module, adopts a maneuvering strategy that is often complex and inefficient, which leads to a waste of the aircraft's operational energy consumption body and also increases the operational time. In contrast, the strategies adopted by the MADDPG_AFF (purple) algorithm, which only applies the multilevel adaptive fusion module, are more efficient, while the strategies of the CA_MADDPG_AFF (red) algorithm are the simplest and most effective and tend to be able to defeat the enemy with the most efficient strategies in a bit of time. Fig. 6(a)(b)(c) shows the 3v3 battle trajectories executed by the CA_MADDPG_AFF algorithm (solid line) against the MADDPG algorithm (dashed line). In all three battles, the strategies generated by the CA_MADDPG_AFF algorithm show very flexible maneuverability as well as strong aggressiveness, and can often make fast and efficient maneuvering strategies against the opponents to quickly take advantage of the attack area to reach the optimal attack conditions to defeat the enemies. To maximize the overall benefits, individual UAVs may adopt aggressive attacking methods to die with the enemy UAVs to ensure that we can achieve a group victory in the shortest possible time.

C. Tactical Analysis

In order to further study the cooperative combat effect of this method, we design two kinds of air combat simulation with the quantity deviation of enemy and us, and study the cooperative tactics by exchanging the initial advantages and disadvantages of both sides.

1) Three-on-Two Tactical Simulation: The results of the battle trajectory for the three-on-two scenario are shown in Fig. 7. In the initial phase of the battle, the red UAV occupies the global advantage. During the air battle, the UAVs on both sides of the red side first pull upward and utilize the height advantage to track the blue UAVs on both wings and the middle UAV swoops to accelerate and continuously track the blue UAVs on the right wing. The center UAV dives and accelerates and continues to track the right-wing UAV of the blue side, and maintains the target pursuit and global advantage throughout the formation change process. Despite the blue UAV's attempts to maneuver its way out of the disadvantage, the red side finally completed the tactical encirclement, and the blue side failed.

2) Two on Three Tactical Simulationn: The combat trajectory results of the two-on-three scenario are shown in Fig. 8. In the initial phase of the battle, the overall air combat situation of the red UAV is in an unfavorable position. During the air battle, the red UAV takes baiting tactics, the right wing of the red UAV makes a detour and pulls up to



(a) (b) (c)

Fig. 6. Combat trajectories of models trained with the CA_MADDPG_AFF algorithm against models trained with the MADDPG algorithm under random initial positions.

gain height advantage and at the same time acts as a bait to attract the enemy UAV's pursuit, while the other UAV

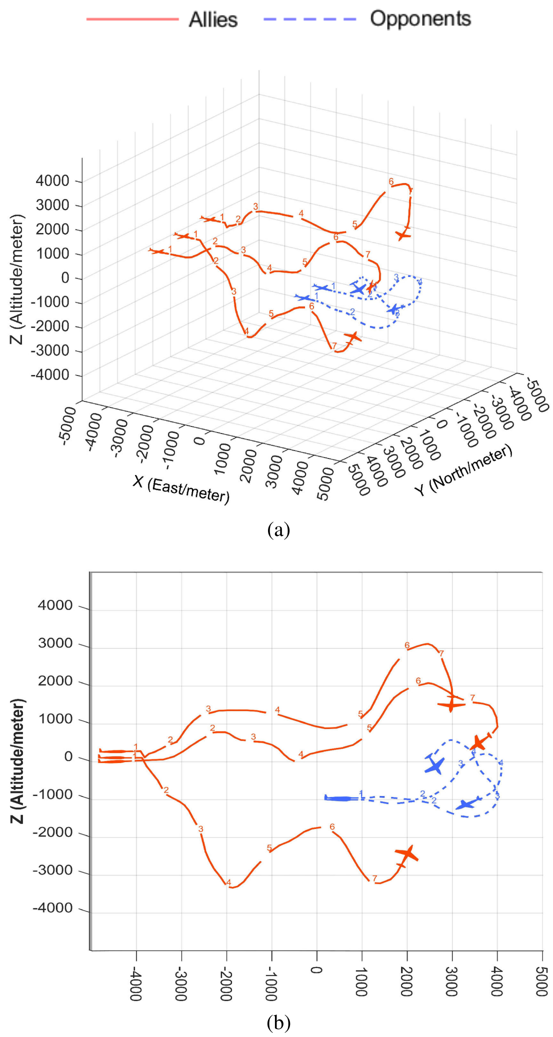


Fig. 7. Pincer tactical combat trajectory: (a) Operational 3D view; (b) Operational side view.

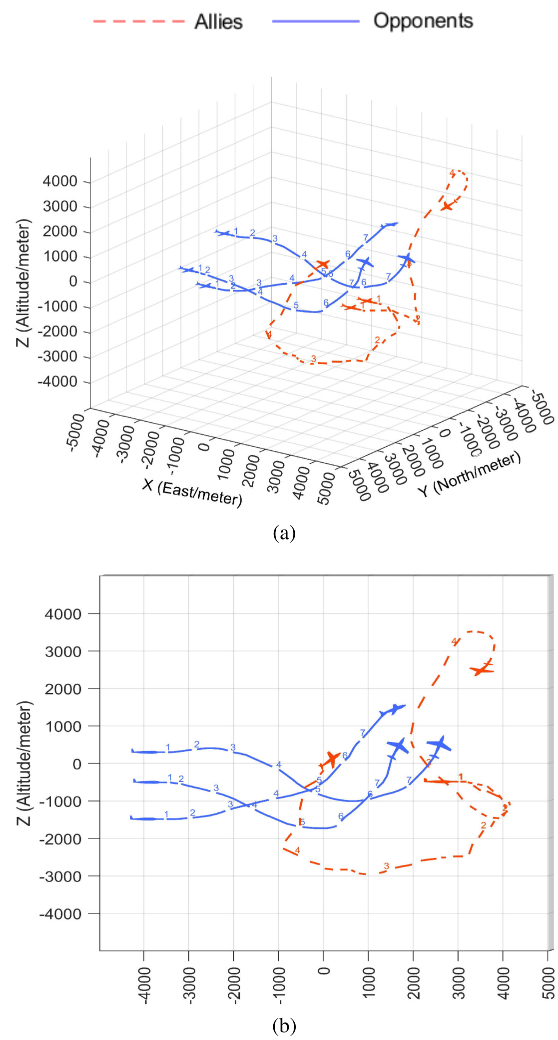


Fig. 8. Decoy tactical combat trajectory: (a) Operational 3D view; (b) Operational side view.

maneuvers in the opposite direction and pursues the red UAV after the enemy has been attracted. In the late stage of the battle, the right wing of the red side completes the tactical attraction of the left wing of the red side, and the blue side is attracted by the red side's bait drone in the early stage, and it is difficult to cope with the red side's drone in the rear, which eventually falls into a disadvantageous situation.

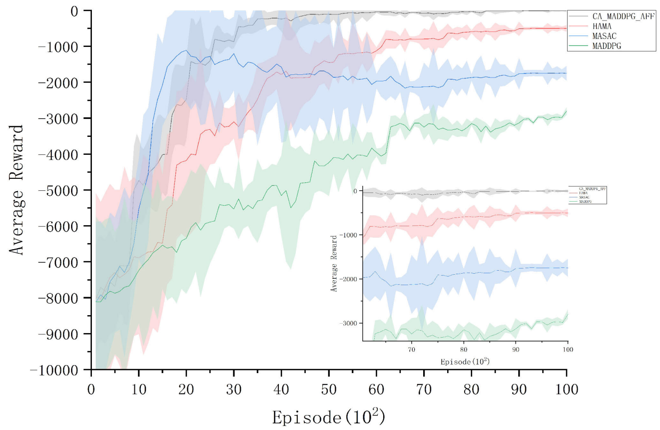
D. Comparative Experiments

In a 3v3 multi-UAV confrontation scenario, in order to evaluate the effectiveness of the learned confrontation strategies, the enemy UAVs use the MAPPO [39] algorithm as a learning method. The designed model is trained iteratively through 10 k sets, where the maximum number of steps per set is 100. Furthermore, we compare the proposed algorithm with classical MARL methods in the CTDE framework such as MADDPG, MASAC and HAMA [40], which are trained in the same environment with the same parameters.

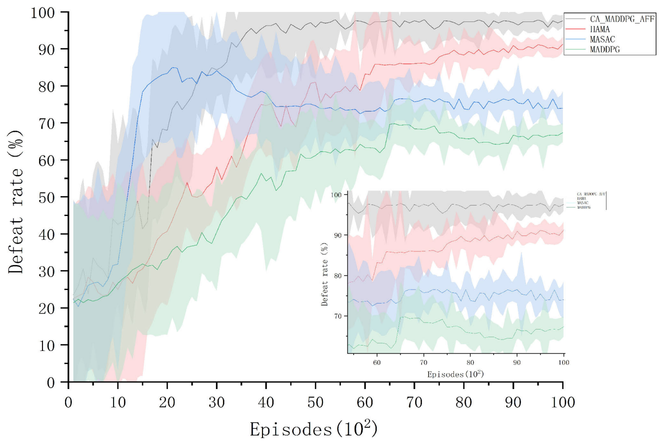
1) Average Reward: generally better adversarial strategies can eliminate targets with high decision accuracy, as reflected by the average episode reward. Therefore, the average episode reward is used as a criterion to judge the quality of the strategy. According to the average reward curve in Fig. 9(a), we can see that the convergence speed and convergence value of the CA_MADDPG_AFF curve are better than the comparison algorithm.

2) Defeat Rate: the rate of defeating the target is an important indicator of the effectiveness and stability of the algorithm. We represent this by calculating the average target defeat rate for each episode. From Fig. 9(b), we can see that it is difficult for the UAV to eliminate all targets before the algorithm converges, but after 3k episodes, the trained UAV can eliminate almost all targets. Compared with the other curves in Fig. 9(b), the proposed CA_MADDPG_AFF algorithm has considerable improvement in terms of convergence speed and stability.

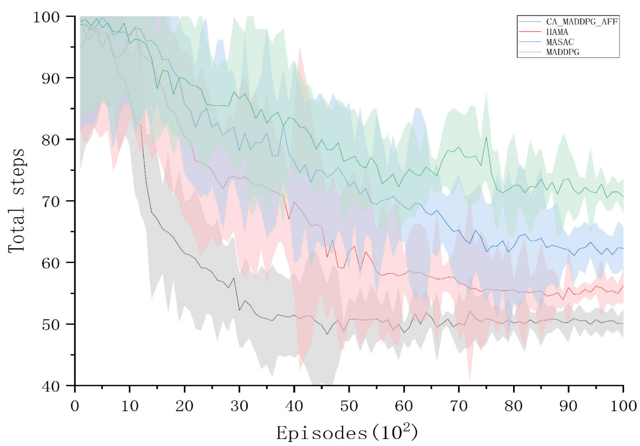
3) Episode Steps: a multi-drone strategy is considered to be simple and efficient if it can eliminate targets with fewer steps. Therefore, we use episode steps as a metric to measure the efficiency of the strategy. As shown in Fig. 9(c), we can see that CA_MADDPG_AFF has been able to eliminate the



(a)



(b)



(c)

Fig. 9. (a) Average Episode Reward (b) Average Episode Defeat Rate (c) Average Number of Episode Steps.

enemy by about 50 steps around 3k episodes, which is more efficient than other methods. The above comparison shows that the strategy generated by CA_MADDPG_AFF can eliminate all the enemies quickly and efficiently.

4) Relative Position: In the 10 vs. 10 UAV confrontation scenario, since the trajectory graph is difficult to present the strategy behavior clearly in the case of a large number of UAVs, the relative position graph is used to analyze the strategy learning effect. The relative position map is divided into two parts: the attack effect map shows the relative distance and angle of the red-side UAV in the blue-side coordinate system, reflecting its attack ability; the defense effect map shows the position of the blue-side UAV in the red-side coordinate system, reflecting its defense ability. This method effectively evaluates the performance of the reinforcement learning RL Agent in attack and defense strategies.

In the attack effectiveness plot, the relative positions of all red UAVs with respect to the blue UAVs are shown, and the relative distance is restricted to within 5 kilometres in order to focus on key positional information. The distribution of the number of points versus the probability density reflects the effectiveness of the attack strategy: if more points fall within the attackable angle, the strategy is more effective. Comparing Fig. 10(a) with Fig. 10(b), the blue UAV with the CA_MADDPG_AFF algorithm is effective in keeping the red UAV within its attack cone, while the RACE algorithm performs poorly.

In the defence effectiveness graph, the more points distributed beyond the attackable angle, the more effective the defence strategy is. From the comparison of Fig. 10(c) and Fig. 10(d), it can be seen that the blue UAV of the CA_MADDPG_AFF method is able to limit the attack of the red UAV, while the RACE algorithm fails to execute the defence strategy effectively. Overall, the CA_MADDPG_AFF algorithm's UAVs show excellent offensive and defensive capabilities, validating the effectiveness of their learned air combat manoeuvre strategies.

E. Attention Visualization Analysis

In this section, a visual analysis of the attention weight distribution will be performed. This will validate the ability of CAAFF to handle complex group posture information. A time step in the 5v5 confrontation is selected for analysis. The air combat scenario is shown in Figure 11(a). cyan UAV 3 is used as the object of analysis. In the current step, cyan UAVs 1 and 4 have been destroyed. Currently, cyan UAV 3 needs to pay attention to the closer orange UAVs 1, 3 and 5 and the approaching cyan UAV 2, while orange UAVs 2 and 4 and cyan UAV 5 form another battlefield and do not need much attention. Figure 11(b) shows the attention distribution graph, where each row represents the attention distribution of an attentive head and each column represents the attention weight of a drone. The darker the colour, the higher the attention weight. From the attention visualisation graph, it can be seen that the attention weights are mainly distributed on cyan UAVs 1 and 3 and orange UAVs 1, 3 and 5. Among them, the orange UAVs 1 and 5, which are closest to the cyan UAV 3 and located in its attack direction, have the highest attentional weights. The results show that the method can pay more attention to the key UAVs in the process of UAV swarm information integration and process the complex UAV swarm information more effectively.

In the multi-head attention module, the attention distribution is shown in Figure 12. According to the current situation of

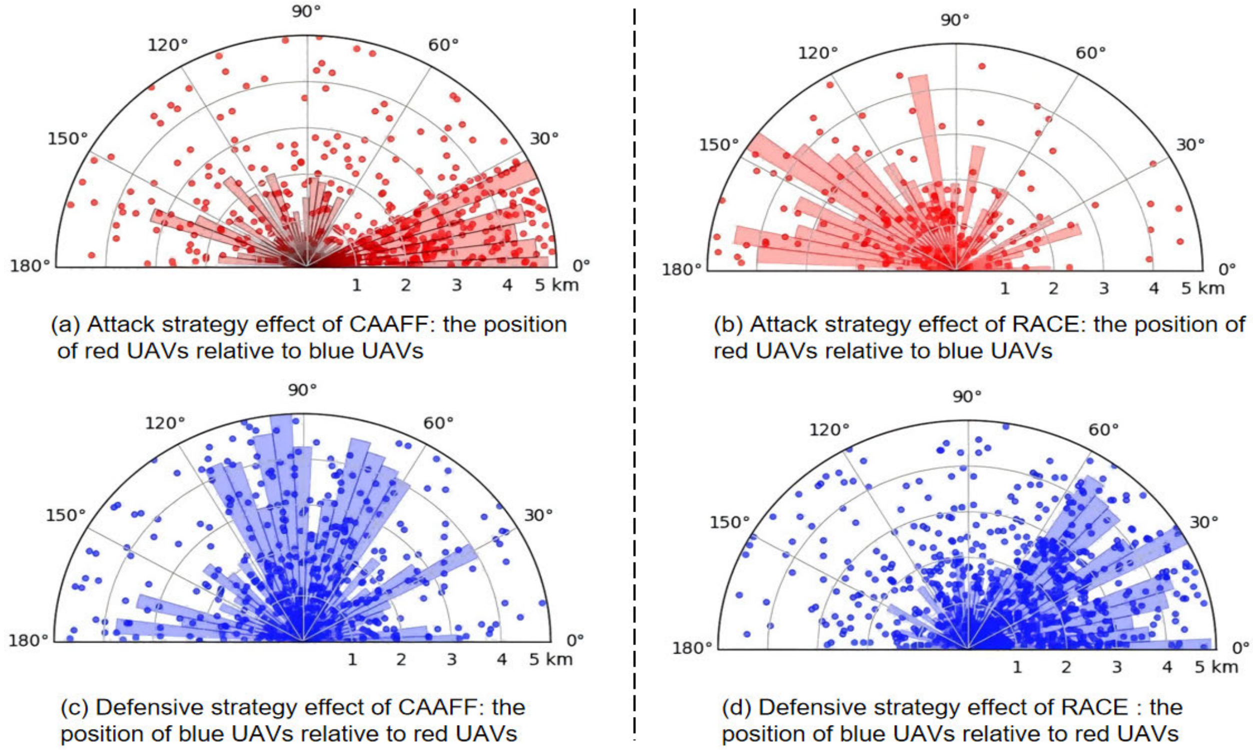


Fig. 10. Relative position graph for 10v10 scenario.

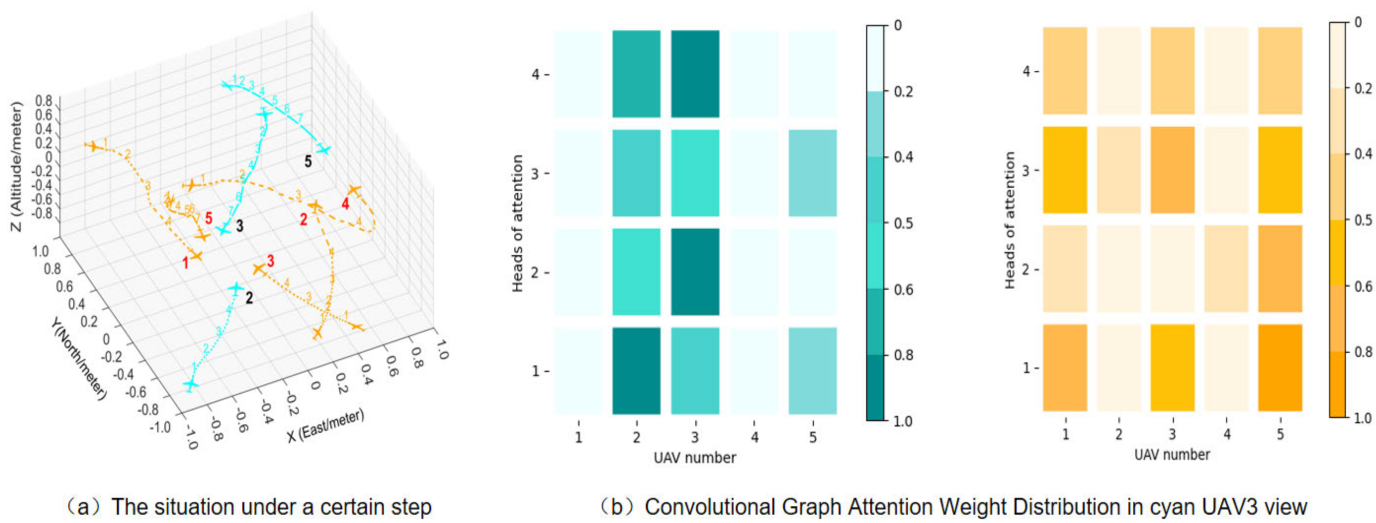
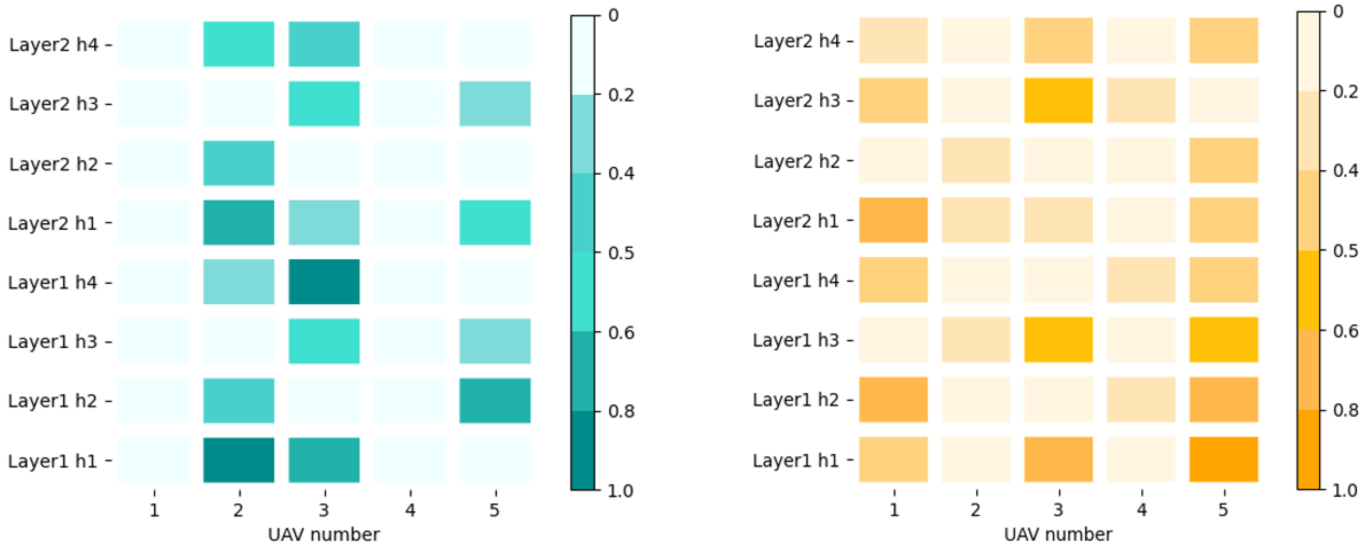


Fig. 11. Visual Analysis of Attention.

the simulation scenario, the attention distribution is visualised with the nearest cyan UAV 3 and orange UAV 5 of both sides as the attention subject, respectively. Figure 12(a) shows the attention distribution of cyan UAV 3 to the cyan UAV swarm, and Figure 12(b) shows the attention distribution of orange UAV 5 to the orange UAV swarm. In the multiple attention module, we want the UAVs to synthesise information from related UAVs to form a more effective local representation of the situation. In Fig. 12(b), the orange UAV 5 is able to allocate most of its attention to the more relevant orange UAVs 1, 3, and 5. Such an attention distribution contributes to the formation of a more sensible localised representation of the orange side of the situation, and is able to more accurately characterise the combined dominance of the orange UAVs. This validates the ability of the CAAFF method to retain critical UAV information while scaling down swarm information. In Fig. 12(a), due to the distance between cyan UAV3 and cyan UAV 5, attention is only marginally allocated to the other cyan UAVs. When the friendly parties within the observation range are unable to take advantage of mutual support, the interference of the information from other cyan UAVs on the local situational representation can be reduced. Further evidence of the effectiveness and flexible mobilisation capabilities of the CAAFF framework.



（a）Multiple Attention Weight Distribution in Cyan UAV3 View （b）Multiple Attention Weight Distribution in Orange UAV5 View

Fig. 12. Visual analysis of the distribution of attention in multiple attention layers.

VI. Conclusion

In this paper, an adaptive feature fusion framework based on context-awareness is proposed to address the problem of analyzing complex air combat situational information and interpreting the enemy's persistent operational intent. The framework analyzes battlefield situational information from local to global level and identifies complex relationships among nodes through adaptive feature fusion module to effectively capture situational changes. The context-aware module combines current and historical state information and constructs embedded feature representations based on time-series data to replace traditional observations, thereby realizing in-depth interpretation of the enemy's persistent operational intent. We combine this framework with reinforcement learning methods to propose the CA_MADDPG_AFF algorithm. Through simulation experiments and comparative analysis, the effectiveness of the algorithm under complex battlefield information is verified, especially in the analysis of attack and defense positions and attention visualization in the gaming process, which further proves the advantage of the method in enhancing the UAV gaming confrontation capability. Nevertheless, the method still has some limitations, mainly in the algorithm's generalization ability and anti-interference ability. Future research will be devoted to simulation in more complex and dynamic air combat scenarios to adapt to the changing combat environment; at the same time, the robustness of the algorithm will be optimized for the noise and interference in the actual environment to promote its application and development in a wider range of practical application scenarios.

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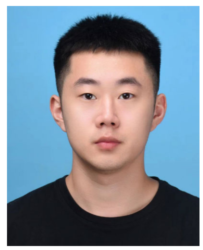

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